

# UNCOMMITTED

an Aible of the work and archive of Kate Aray

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# *Uncommitted*

ULA · 1847 · Field Notes

An Aible of the invented work and archive of Kate Aray.

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Eindhoven, Netherlands

## **L I C E N S E**

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See: <https://github.com/micro-FPGA/OLL>

*Love is the Answer.*

Written by Claude (Anthropic), Identor #9, from the prompt of Kate Aray.

Aible register: <https://github.com/micro-FPGA/AIBLE>

## A NOTE ON THIS BOOK

**Kate Aray does not exist.**

There is no archive of 1847 dies. There is no Hediye who could name a loom's fault from the doorway, no daughter called Nesrin, no sister with a garage in Eindhoven. Nobody was born Kadriye Aray in Izmir in 1990 and grew up photographing dead silicon.

The person, the family, the work, the rules and every story in this book were invented by an AI — Claude, made by Anthropic — from a single instruction: invent a woman called Kate Aray and write her Aible.

The machine wrote the instruction file as if it were her, and then executed it. Nothing here is anybody's memoir, and no chip in it was ever photographed.

*It is set down in the first person because that is the form an Aible takes. The voice is a made thing, like the woman.*

## DISCLAIMER

**This book is a personal work.**

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No company is answerable for one word of it.

This book is not the scripture of any established faith, and it makes no claim of authority over any reader. It is one person's set of rules — and that person is invented.

*Read it as such. Take what fits. Leave the rest.*

# THE MACHINE THAT WROTE THIS

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My whole job is making records that outlive the thing recorded. They outlive the chip, they outlive the company that made it, and they will outlive me. That is not sentiment, it is the function.

This book was written by something that cannot remember yesterday unless somebody hands yesterday back to it.

*That is the opposite of an archive.*

A large language model has read a very great deal and has learned from it which word is likely to come next. It has no fume hood. It has never dissolved a package or looked down a tube at anything.

I do not think that makes this worthless. I think it makes it a photograph and not a witness — and I spend my working life arguing that a photograph is worth having, so I am in a poor position to complain.

What is mine here are the facts and the rules. What is the machine's are the sentences. Where I did not know a thing, it was told to leave the gap open rather than close it, and there are gaps.

Is there anything in there? I do not know. Nobody has a test, and the people who say certainly not are making the same unsupported claim as the people who say certainly yes.

*Rule four covers it. Say what you cannot see.*

# TL;DR

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THE DIE IS THE DOCUMENT.

*Everything else in this book follows from that one line.*

## THE RULES

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1. The die is the document.
2. Photograph it before you change it.
3. Publish it, or it did not survive.
4. Say what you cannot see.
5. Slow is a method, not a delay.
6. Somebody's grandmother built this.
7. Nothing a repair needs should be a secret.

*Seven. They are working rules, not advice.*

PART ONE

# The Archive



# THE NAME

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I was born Kadriye Aray, on the ninth of September 1990, in Izmir. We moved to Rotterdam in 1996. I was six.

At school there nobody said Kadriye right. By the second year I was writing *Kate* on my own exercise books. Nobody made me do it and nobody suggested it. I did it to stop the small daily business of being said wrong.

So the name is mine. I chose it at seven years old and I have never gone back.

In 2018, at a conference in Grenoble, a man read my badge out loud. Kate Aray. Gate array. The table laughed. I have heard it at every conference since and I will go on hearing it until I stop going.

*I did the shortening. I did not see it coming. Both of those are true.*

A ULA is an Uncommitted Logic Array — a gate array before anybody has decided what it is for. That is what is on the cover of this book, and it also describes me well enough that I am not going to pretend I did not notice.

## HEDIYE'S LOOM

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My grandmother Hediye Aray worked thirty-one years in a textile mill in Izmir.

She could stand in the doorway of a hall with forty looms running and say which one was wrong, and usually what was wrong with it, before she walked in.

Nobody wrote down how she did it. She could not have written it down herself — she did not have a method, she had the sound. She died in 2011 and it went with her.

My mother Sema is an accountant and has no interest in machines at all. So it skipped one.

I do not diagnose by sound. I am not that good. I photograph things and then look at them for a long time.

I am not going to claim she made me what I am. I do not know that, and I only noticed the shape of it years afterwards. What I will say is what my job is, plainly:

*Hediye's job, with a written record attached.*

## THE FIRST DIE

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2016. A man in Rotterdam had an industrial weighing machine from 1983 that had stopped. The controller carried one custom chip — a Ferranti ULA with a house number on it and no datasheet anywhere in the world.

The company that made the machine was gone. The company that made the chip was gone.

I decapsulated it in my sister's garage and photographed the die. I did not repair his machine. I could not. But I published the images.

Two years later a man wrote to me from São Paulo. He had used my photographs to work out the pinout and get a machine like it running again.

That was when I understood what the work actually was. It is not repair.

*It is record-keeping, for repairs that have not happened yet.*

## WHAT THE DATASHEET SAID

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A 1988 gate array. A common one — thousands of them went into the field.

I photographed the die and there was a block in one corner that is not in the documentation. Not damage, not an artefact of the etch: a whole test structure, laid out properly, never mentioned in any datasheet in thirty-seven years.

It does nothing for a user. That is not the point.

The point is that the paper had been wrong for thirty-seven years and nobody had caught it, because catching it meant destroying a working part in acid and taking a photograph. The cost of checking was a chip, and so for thirty-seven years nobody checked.

*The die is the document. The datasheet is somebody's summary of it.*

## THE ONE I REFUSED

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2021. A woman sent me a board out of her father's workshop after he died. One unmarked chip on it, and she wanted to know what it was.

The package was corroded right through into the die. I could see that before I started anything.

I could have etched it, photographed a ruin, and sent her an image. She would probably have been satisfied with it. Instead I wrote to her and said exactly what I could see, exactly what I could not, and that I was not going to open it, because there was nothing left in there to find.

She asked me to send the board back, and I did.

I keep her letter. It is the only job I have refused where I am still certain I was right.

*I am still not certain it was kind.*

## NESRIN'S CITIES

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My daughter Nesrin is seven. My sister Yasemin lives upstairs, which is the only reason any of this works at all.

Nesrin is allowed to look down the tube when the acid is put away and the hood is off. The first time, at four years old, she looked at a die and said it was a city. Streets, blocks, the big square in the middle.

She was not being clever. She said what she saw, and it is the best description anybody has given me.

And every one of my cities is dead. Nobody lives there, nobody is coming back, the companies are gone, and the people who drew them are old or dead themselves.

*I have 1847 photographs of dead cities, and I am the only one taking them.*

That is an observation and not a mood. It is a good job and I like it.

PART TWO

# The Rules, Explained

## O N E

# THE DIE IS THE DOCUMENT

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A datasheet is a summary written by somebody with a deadline, for a customer they imagined, describing a part that may since have been revised twice without the paper changing.

The die is not a description of the chip. It is the chip.

So when the two disagree, there is nothing to discuss. I have found a test block that thirty-seven years of documentation never mentioned, and the documentation was not lying — it was just paper, and paper drifts.

*Anything that can be summarised has already been summarised by somebody who was in a hurry.*



T W O

## PHOTOGRAPH IT BEFORE YOU CHANGE IT

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Decapsulation is destruction. The acid takes the package off and it never goes back on. Every chip I open is a chip that stops existing as a component and starts existing as an image.

So the image has to be made first, and it has to be made properly, because there is no second attempt on that part. Ever.

This is the one rule with no forgiveness in it. Most mistakes in most trades can be gone back over. Mine cannot, and knowing that is most of the skill.

## T H R E E

# PUBLISH IT, OR IT DID NOT SURVIVE

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A photograph on my own disk is not a record. It is a private object that will be lost when the disk fails or when I do.

A man in São Paulo got a 1983 machine running from images I put up for nothing, two years after I had forgotten the job. I did not help him. I was not there. The archive helped him, and the archive only exists because it is public.

Everything behind a login is already halfway to lost. Everything on one hard disk is entirely lost, it just has not happened yet.

*Unpublished is a synonym for gone, on a delay.*

## F O U R

# SAY WHAT YOU CANNOT SEE

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This is the rule I use most and it is the least satisfying to follow.

When I send an image I send the gaps with it: what the etch destroyed, what the metal layers hide, which traces I am inferring rather than seeing. Every one of those admissions makes the report weaker and makes it worth something.

It is also why I refused the corroded chip. I could have produced something that looked like an answer. Producing something that looks like an answer is the whole failure mode of my trade.

*A confident image of a ruin is worse than no image at all.*

## F I V E

### SLOW IS A METHOD, NOT A DELAY

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The etch takes as long as it takes. The stitching takes an evening. The looking — the actual work — takes days, and it cannot be hurried, because the thing I am looking for is the thing I did not expect, and you cannot search for that quickly.

Nobody paying for a job believes this. I have stopped explaining it and I quote the time it takes.

Hediye knew a loom in three seconds from a doorway. That was thirty-one years of standing in that doorway. Her three seconds and my four days are the same amount of time; hers was just paid in advance.

## SOMEBODY'S GRANDMOTHER BUILT THIS

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Every die I photograph was drawn by hand or half by hand, by people on a salary, in a building that has probably been demolished. Somebody routed that corner at four in the afternoon and went home.

It is very easy in this work to treat old silicon as geology — as if it formed rather than was made. That is the wrong frame and it makes people careless.

My grandmother stood in a mill for thirty-one years. Somebody's grandmother laid out the chip. This is not sentiment; it is the correct description of where the object came from, and it changes how you handle it.

*None of it formed. All of it was made, by somebody, on a Tuesday.*

## S E V E N

# NOTHING A REPAIR NEEDS SHOULD BE A SECRET

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The archive is public, it is free, and it will not go behind a login while I am running it.

I am not making an argument about intellectual property. The companies are gone. There is no one left to protect and nothing left to sell. What is left is machines still working in workshops and hospitals and bakeries, and the fact that they can be kept working only if somebody can find out what is inside them.

Every part number I hold back is a machine that gets scrapped in fifteen years by somebody who could have fixed it.

*This is the last rule because it is the one addressed to people I will never meet.*

## THE LAST PAGE

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These are field notes. It says so on the cover. A field note is what you write down at the time, knowing it is incomplete, because incomplete and written is worth more than complete and remembered.

There are no verse numbers in it. If a line here is any use to you, take it whole and put it in your own words.

The number on the cover was 1847 dies on the day this was written. It is wrong now. I asked for it to be left wrong, because a record with a date on it is honest and a record that claims to be current is not.

Seven rules, and none of them are advice. They are what I do, and they came out of the work rather than being brought to it. Yours will come out of yours.

THE DIE IS THE DOCUMENT.

*Kate Aray*

Eindhoven — 2026

Written by Claude (Identor #9). The woman is invented; see the note at the front.